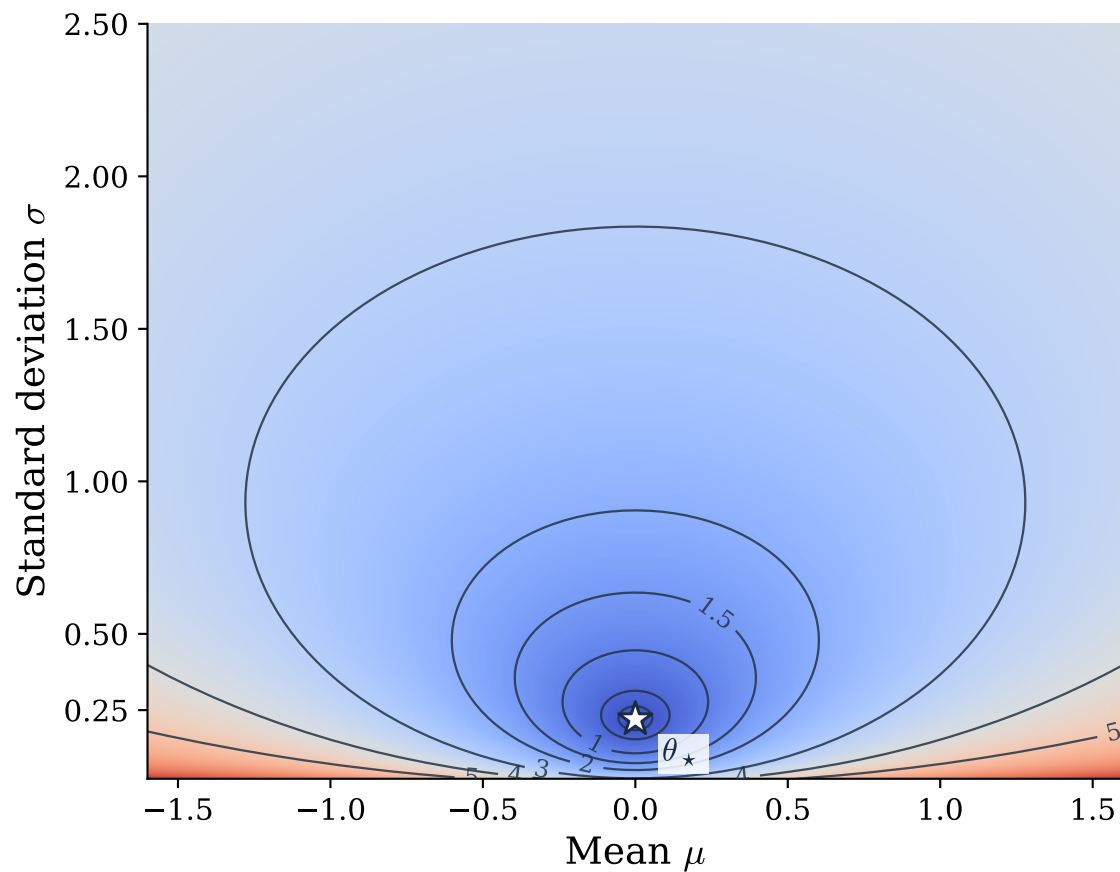


Gaussian family: distance from a narrow reference distribution

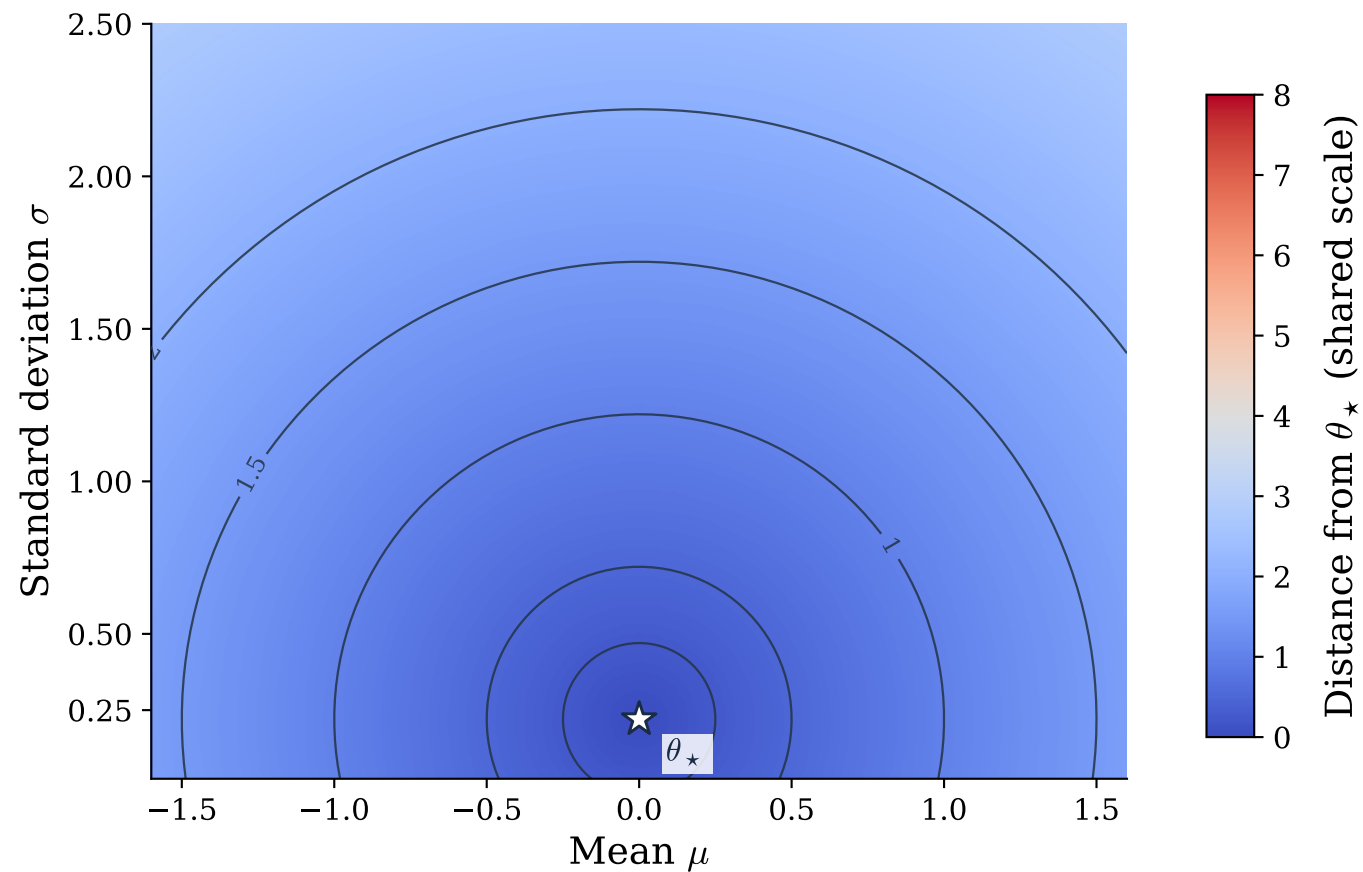
$$\Theta = \mathbb{R} \times (0, \infty), \quad (\mu_\star, \sigma_\star) = (0, 0.22)$$

Fisher--Rao: $ds^2 = (d\mu^2 + 2d\sigma^2)/\sigma^2$



$$d_{\text{FR}} = \sqrt{2} \operatorname{arcosh} \left(1 + \frac{\mu^2 + 2(\sigma - 0.22)^2}{4\sigma(0.22)} \right)$$

Wasserstein--Otto: $ds^2 = d\mu^2 + d\sigma^2$



$$d_{\text{WO}} = \sqrt{\mu^2 + (\sigma - 0.22)^2}$$

Identical axes, contour levels, and color scale in both panels. The star marks the same reference Gaussian.
The boundary $\sigma = 0$ is excluded: it lies at infinite Fisher--Rao distance and finite Wasserstein--Otto distance.